

Bayes decision theory

Introduction

- In this lecture we will cover the following topics:
 - Bayesian decision theory
 - The naïve Bayes assumption
 - Bayes classifier minimizes probability of error
 - Risk minimization
 - Discriminant functions and decision surfaces

Spam

- It's the early 2000, and spam emails offering you to buy Viagra are flourishing.
- Cannot just filter on keywords.
- Mathematically, we want to find $P(\text{spam}|\text{content})$.



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Payments. 90,000+ Satisfied Customers!

Top Selling 100% Quality & Satisfaction guaranteed!

Tokenization and learning

- Break email into words W_i
- Discrete events: $P(W_i) \geq 0$
- Notation
 - S : Spam (illegitimate messages)
 - H : Ham (legitimate messages)
- How to get an expression for $P(\text{spam}|\text{content})$?
- Bayes' rule:
 - $P(S|W_i)$
$$= \frac{P(W_i|S)P(S)}{P(W_i)}$$
- where:
 - $P(W_i) = P(W_i|S)P(S) + P(W_i|H)P(H)$

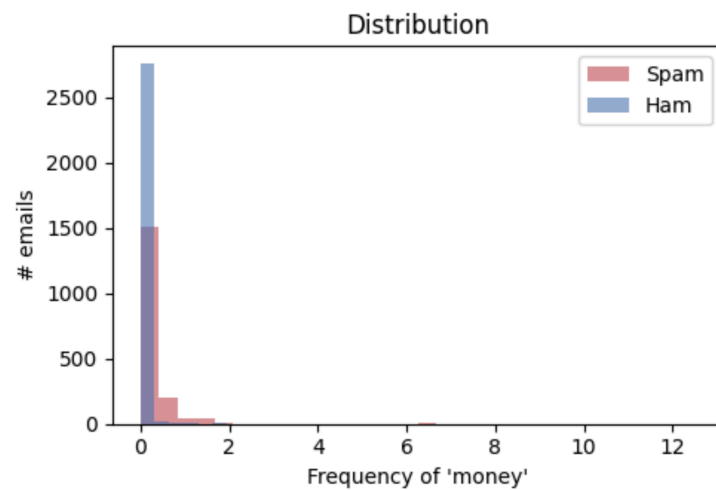
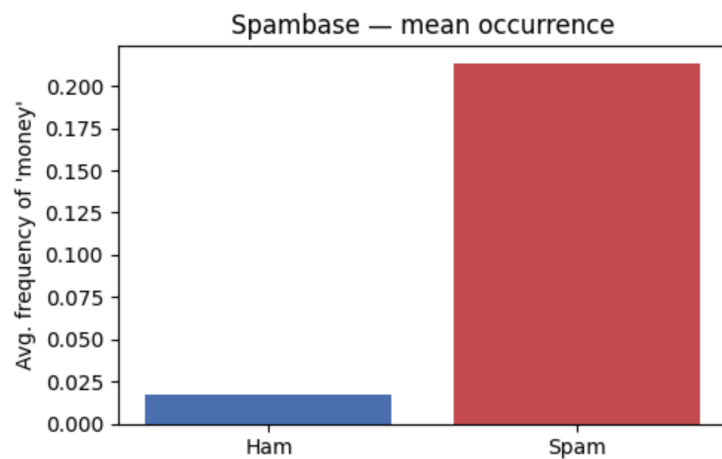
How to learn from data

- Collect Spam and Ham and learn (estimate) $P(W_i|S)$ and $P(W_i|H)$!
- Example: $W_i = \text{"Money"}$
- Let us look at the data — pick which word to inspect below.

money

george

Spam mean: 0.213 Ham mean: 0.017



Detection

- Need $P(W_1 \cap W_2 \cap \dots \cap W_K | S)$
- Difficult!
 - Think about what this term is actually saying.
- Need to simplify.
- Note: modern large language models are modeling something very similar to the expression above.

3.1 Unsupervised pre-training

Given an unsupervised corpus of tokens $\mathcal{U} = \{u_1, \dots, u_n\}$, we use a standard language modeling objective to maximize the following likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

where k is the size of the context window, and the conditional probability P is modeled using a neural network with parameters Θ . These parameters are trained using stochastic gradient descent [51].

References:

- Radford et al. — *Improving Language Understanding by Generative Pre-Training*, 2018.
- Bengio et al. — *A Neural Probabilistic Language Model*, 2003.

Naïve Bayes assumption

- Assume independence!
- $P(A \cap B)$
 $= P(A)P(B)$
- Then we can compute
 $P(S|W_1 \cap W_2 \cap \dots$
 $\cap W_K)$
:

$$P(S|W_1, \dots, W_K) = \frac{P(S) \prod_i P(W_i|S)}{P(W_1, \dots, W_K)}$$

Bayes classifier

- Shifting focus a bit.
- How can we think more generally about classification using Bayes' rule?
- First: think continuous random variables $\rightarrow$
- Let us start with considering two classes w_1 and w_2 .

Core idea of Bayes classifier

- Given a new sample $\mathbf{x}$, how can we decide if we want to classify it as belonging to class w_1 ?
- What about:
 - $P(w_1|\mathbf{x}) > P(w_2|\mathbf{x})$?
- else class w_2 .

Leveraging Bayes' rule

- $P(w_1|\mathbf{x})$ is not straightforward to model.
- Bayes' rule to the rescue:
 - $P(w_1)p(\mathbf{x}|w_1)$
 $> P(w_2)p(\mathbf{x}|w_2)$
- Much easier to work with $\rightarrow$

Example with balanced dataset

- Consider the case where $P(w_1) = P(w_2) = 0.5$.
- Draw $\rightarrow$

Example with unbalanced dataset

- Consider the case where $P(w_1) = 0.25$ and $P(w_2) = 0.75$.
- Draw $\rightarrow$

Example with overlapping Gaussians

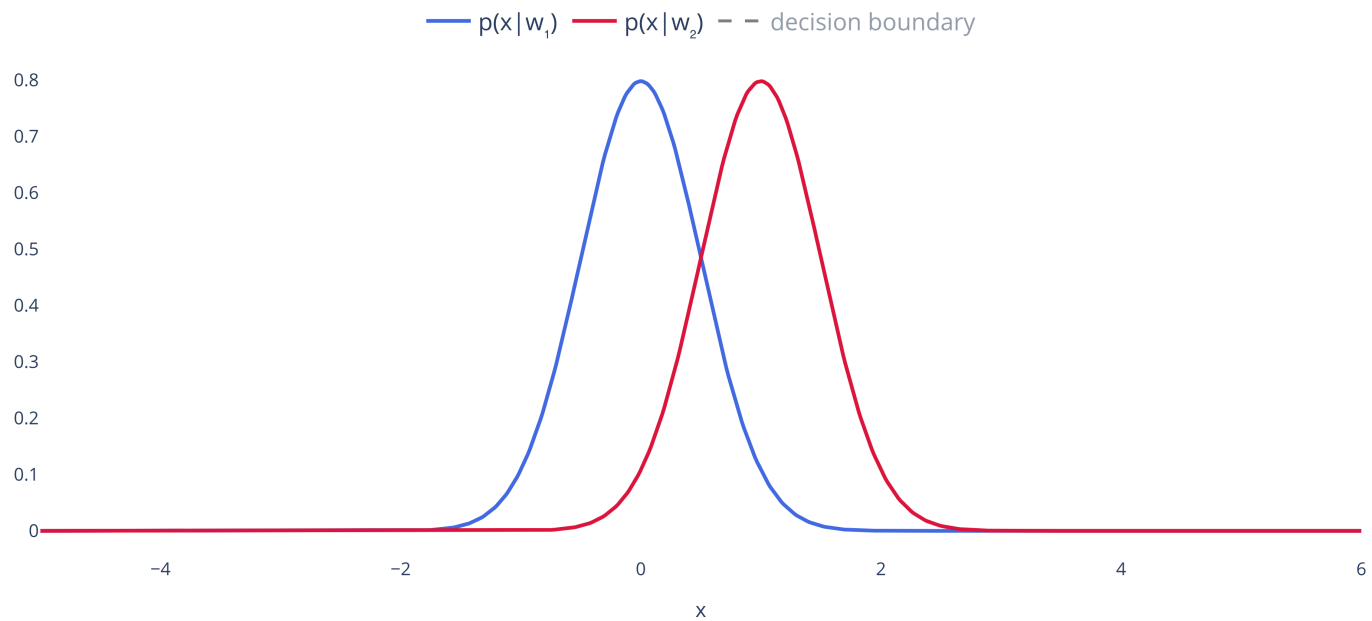
- Each tab is a different parameter setting — read the label to see exactly which parameters are in play. Means, standard deviations, and (in the last tab) the prior can all change.
- Click the "**decision boundary**" entry in the legend to toggle it on or off. It starts **off** so the densities stand alone first; turn it on to see where the Bayes classifier would assign each x .

$\mu_1=0, \mu_2=1, \sigma_1=\sigma_2=0.5$

$\mu_1=0, \mu_2=2, \sigma_1=\sigma_2=0.5$

$\mu_1=0, \mu_2=1, \sigma_1=0.5, \sigma_2=0.6$

$\mu_1=0, \mu_2=1, \sigma=0.5, P(w_1)=0.1$



Probability of error

- Want:
 - Algorithm that can solve task with high performance.
 - Think spam filter that removes all spam and keeps all ham.
- Problem:
 - Real data is complicated, we will make mistakes!
 - What is the probability of making an error with a Bayes classifier?
- The probability of error can be modelled as follows:

$$\begin{aligned} P_e &= P(\mathbf{x} \in R_1, w_2) + P(\mathbf{x} \in R_2, w_1) \\ &= \int_{R_1} p(\mathbf{x}|w_2)P(w_2) d\mathbf{x} + \int_{R_2} p(\mathbf{x}|w_1)P(w_1) d\mathbf{x} \end{aligned}$$

Bayes classifier minimizes P_e

- Note that P_e can also be expressed in terms of an integral:

$$P_e = \int_{R_1} P(w_2)p(\mathbf{x}|w_2) d\mathbf{x} + \int_{R_2} P(w_1)p(\mathbf{x}|w_1) d\mathbf{x}$$

- Visual proof $\rightarrow$
- Formal proof as extra exercise.

Going beyond the two-class setting

- Given a new sample $\mathbf{x}$, how can we decide if we want to classify it as belonging to class w_i ?
- $P(w_i)p(\mathbf{x}|w_i) > P(w_j)p(\mathbf{x}|w_j)$ for all $j \neq i$.
- Assign $\mathbf{x}$ to the largest $P(w_i)p(\mathbf{x}|w_i)$.

Dealing with risk in the Bayes classifier

- In many cases, certain types of errors have bigger consequences than others.
- **Example 1:**
 - A spam email getting through the spam filter?
 - Annoying, but manageable.
 - A legitimate email with an important message getting caught in the filter?
 - Potentially very damaging.
- **Example 2:**
 - A benign tumor being classified as malignant in breast cancer screening?
 - Not good; increases workload and chance of unnecessary treatment. But, "better safe than sorry".
 - A malignant tumor being missed?
 - Potentially fatal.
- Can we incorporate this into the classifier?

Introducing risk factors to the Bayes classifier

- Define:
 - λ_{12} : penalty for wrongly classifying w_2 as w_1
 - λ_{21} : penalty for wrongly classifying w_1 as w_2
- Consider the two-class setting again:
 - $\lambda_{12}P(w_1)p(\mathbf{x}|w_1)$
 $> \lambda_{21}P(w_2)p(\mathbf{x}|w_2)$
- Rewrite as a ratio:

$$\frac{p(\mathbf{x}|w_1)}{p(\mathbf{x}|w_2)} > \frac{\lambda_{21}P(w_2)}{\lambda_{12}P(w_1)}$$

- Need to know class-conditional distributions.

Investigating the simplest case — one-dimensional normal distributions

- Let $p(\mathbf{x}|w_i)$ follow a normal probability density function.
- One-dimensional case:

$$p(x|w_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right)$$

- General (multivariate) case:

$$p(\mathbf{x}|w_i) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_i|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\right)$$

Example with two-class problem

- Consider a balanced dataset with class-conditional distributions $p(\mathbf{x}|w_1) \sim \mathcal{N}(\mu_1, \sigma)$ and $p(\mathbf{x}|w_2) \sim \mathcal{N}(\mu_2, \sigma)$.
- Let $P(w_1) = P(w_2)$ — balanced dataset (equiprobable classes).
- Expression for the ratio:

$$\frac{p(\mathbf{x}|w_1)}{p(\mathbf{x}|w_2)} > P(w_2)/P(w_1)$$

Discriminant functions

- Take a step back:
 - What is actually going on?
- Input space is partitioned into regions. Two regions are separated by a decision surface.
- Often mathematically convenient to describe the decision surface using a *discriminant function*:
 - $g_i(\mathbf{x}) = f(P(w_i)p(\mathbf{x}|w_i))$

Example for Gaussian case

- Revisiting the Gaussian likelihood.
- What could be a suitable choice for f ?
- In this case, $g_i(\mathbf{x})$ becomes:

$$g_i(\mathbf{x})$$

Decision surfaces

- Decision surface is where the discriminant functions are equal:
 - $g_{ij}(\mathbf{x}) = g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0$
- Quadratic surface! Draw example $\rightarrow$
- Often we assume equal covariance structure. What happens then?

Continue example for Gaussian case

- Remember:

$$g_{ij}(\mathbf{x})$$

$$= g_i(\mathbf{x})$$

$$- g_j(\mathbf{x})$$

$$= 0$$

- 1st term $\rightarrow$
- 2nd term $\rightarrow$

Special case for equiprobable classes with the same covariance matrix

- Assume $\Sigma = \sigma \mathbf{I}$
- Then the discriminant function $g_{ij}(\mathbf{x})$ becomes $\rightarrow$
- Draw example $\rightarrow$